Spatio-Temporal Denoising of Calcium Imaging Data: Towards Robust Reconstruction for Neural Activity Analysis

Aman Aggarwal[1], Kadiyala Suraj Kumar[1], Siddhartha Ravva[1]
[1] Department of Computer Science and Engineering, Amrita School of Computing, Amrita Vishwa Vidyapeetham, Bengaluru, India
[1]bl.en.u4cse23206@bl.students.amrita.edu
[2]bl.en.u4cse23228@bl.students.amrita.edu
[3]bl.en.u4cse23246@bl.students.amrita.edu

I. INTRODUCTION

Understanding how vast population of neurons work in doing vastly different tasks forms the basis of modern neuroscience. Calcium Imaging is one of the techniques used, which can record the activity of this vast neural network through calcium fluorescence. Calcium sensitive fluorescent dyes are injected into the living tissue and these act as markers for seeing the neuron activity. These dyes when bind to calcium ions change in their fluorescence properties, this fluorescence is then measured using Two-photon microscope which records high resolution three-dimensional images. These images when captured frequently form the spatiotemporal data from which neuron activity of individual neurons must be extracted.

A fundamental problem that limits utility of calcium imaging is the presence of noise in the data. This noise is majorly Photon shot noise which is the randomness in photon arrival, especially in low power or high-speed conditions. Other is the noise in the equipment used to find this read the data and most significantly structured background contamination from out-of-focus fluorescence. This combination of noise degrades the SNR (signal to noise ratio) and obscures the meaning neural events and corrupts the data. Hence, the accurate identification of neuron pathways used, and the inference neural spike times are compromised.

Initial approaches to this problem relied on traditional image processing filters and statistical methods like Principal Component Analysis, which often blurred the images and discarded valuable information. A significant advantage came with model-based approaches like CNMF (constrained non-negative matrix factorization) which can simultaneously separate neural signals from background and deconvolve spike times. More recently, deep learning has been used, Supervised methods have demonstrated a remarkable performance but require a difficult to acquire ground truth data. While self-supervised methods like DeepCAD which use the temporal structure of the data to denoise without a clean reference, these methods typically involve a single monolithic network to handle both spatial and temporal data simultaneously which may not be optimal in preserving the fine-grained temporal data.

While current deep learning models excel in spatial denoising, the fidelity in the temporal data is still lost. Our approach for spatiotemporal calcium image denoising explicitly decouples the problem into its spatial and temporal components. Our key innovation is the hybrid deep learning architecture which has a convolutional U-net which is used for biomedical image segmentation and restoration, used to denoise each frame spatially. The extracted temporal traces from the spatially-denoised video are then processed by a deep residual network. This design of this temporal module is inspired by the demonstrated success of ResNets in accurately classifying complex, one-dimensional biological time-series data such as ECGs, making it exceptionally well-suited to refining the subtle dynamics of calcium traces.

II. LITERATURE SURVEY

A. Calcium imaging analysis – how far have we come? (2021)

This paper reviews the standard computational pipeline for calcium imaging analysis, breaking it down into four stages: Denoising, Motion Correction, Source Extraction (Classification), and Spike Inference (Quantification). Considers data from diverse experimental contexts, including in vitro and in vivo imaging in behaving animals using technologies like two-photon microscopy and miniscopes. Concludes that while automation has improved significantly,

many state-of-the-art tools are still semi-automated and often tailored to specific experimental niches. Cannot analyse complex structures like dendrites, tracking cells over extended periods

B. Analysis pipelines for calcium imaging data (2019)

A computationally focused review on the algorithms used and an emphasis on **Constrained Nonnegative Matrix Factorization (CNMF)**. The pipeline is described as a sequence of motion correction, source extraction, spike inference, and cross-session registration. This registration. This paper focuses on standard 2-photon and 1-photon microendoscopic data. Argues that the massive scale of modern datasets makes fully automated and reproducible pipelines essential. It demonstrates how model-based approaches like CNMF effectively solve critical problems like separating signals from physically overlapping neurons.

C. Computational processing of neural recordings from calcium imaging data (2019)

A review from the developers of the Suite2P package. It covers the core pipeline but adds a fifth step: **Quality Control**. It introduces objective metrics to evaluate the output of analysis pipelines, such as the "isolation distance" of a neuron's signal from the surrounding background. This paper focuses on large-scale two-photon imaging from the mouse visual cortex. In this paper it is observed that the author wants us to use the simplest effective method to ensure results are interpretable and reliable This paper stresses on not introducing hard to detect biases on the results and to do extensive checks to give the perfect results

D. Simultaneous Denoising, Deconvolution, and Demixing of Calcium Imaging Data (2016)

The main paper which introduced the **Constrained Non-negative Matrix Factorization (CNMF)** framework. It models the raw video as a mathematical product of a **spatial matrix** (neuron shapes) and a **temporal matrix**, plus background. By constraining the temporal components with a biophysical model of calcium indicator kinetics, it can simultaneously denoise, deconvolve spikes, and demix signals. The dataset used is a highly diverse set of data, including in vitro neurons, in vivo mouse cortex, whole-brain zebrafish, and dendritic imaging, proving its wide applicability. CNMF became the new state-of-the-art method, outperforming all the previous methods in separating signals from densely packed and overlapping neurons.

E. Denoising Two-Photon Calcium Imaging Data (2011)

A classic statistical signal processing approach. It models each pixel's time-series as a sum of a predictable, stimulus-driven signal (using **harmonic regression**) and temporally correlated noise (using an **autoregressive model**). The dataset used is Two-photon imaging from the ferret visual cortex during a periodic visual stimulus. The method successfully separated the evoked signal from noise, enhancing image contrast and revealing subcellular dynamics not visible with simple averaging. This paper demonstrates the complexity and limitations of building explicit mathematical models for signal and noise, justifying the shift toward deep learning models that can learn these complex distributions from data automatically.

F. Fast, efficient, and accurate neuro-imaging denoising via supervised deep learning (NIDDL) (2022)

A **supervised deep learning** pipeline named NIDDL, which uses an optimized U-Net architecture. Its key methodological innovation is training on pairs of low-SNR and high-SNR images that are **temporally independent**, which simplifies the otherwise complicated process of perfectly acquiring time-aligned ground-truth videos. Primarily C. elegans imaging, where high-SNR ground truth was obtained by using a higher laser power. NIDDL achieves state-of-the-art accuracy with a model that is significantly smaller (20-30x) and faster (3-4x) than other deep learning methods. It can be trained on a small dataset (~500 image pairs). This is a direct competitor and represents the supervised learning approach.

G. Reinforcing neuron extraction and spike inference in calcium imaging using deep self-supervised denoising (DeepCAD) (2020)

A **self-supervised deep learning** method. DeepCAD's core idea is that noise is independent between video frames, while the biological signal is highly correlated. It trains a 3D U-Net to predict a noisy frame using its neighbouring noisy frames as input. This forces the network to learn the underlying signal structure, eliminating **the need for clean ground-truth data**. Validated on simulated data and real two-photon data from awake mice, including public data with simultaneous electrophysiology for ground-truth spike validation. DeepCAD improves imaging SNR by over 10 times, which dramatically improves the performance of downstream neuron finding (52% more neurons detected with CNMF) and the accuracy of spike inference.

H. Automated denoising software for calcium imaging signals using deep learning (CalDenoise) (2024)

A software package named **CalDenoise** that integrates four pipelines: one traditional image-processing pipeline and three based on **Generative Adversarial Networks (GANs)**. The GANs are trained to generate realistic denoised images from noisy inputs, using different strategies to obtain the ground-truth for training. The dataset used is Calcium imaging of interstitial cells of Cajal (ICCs) in the murine colon. The GAN-based pipelines significantly outperform traditional methods and a standard U-Net at removing complex, periodic noise patterns**.**

I. Automatic diagnosis of the 12-lead ECG using a deep neural network (2020)

A supervised deep learning approach using a **deep Residual Network (ResNet)** to perform multi-label classification of cardiac abnormalities directly from raw 1D electrocardiogram (ECG) waveforms. An extremely large-scale clinical dataset of over 2.3 million labelled ECGs. The trained DNN achieved cardiologist-level performance, outperforming medical residents.

J. EZcalcium: Open-Source Toolbox for Analysis of Calcium Imaging Data (2020)

A **software toolbox** for MATLAB that provides a Graphical User Interface (GUI) for existing state-of-the-art algorithms, namely **NoRMCorre** (motion correction) and **CaImAn** (source extraction and deconvolution). Two-photon data from drosophila and mouse cortex. The toolbox makes powerful, command-line-based algorithms accessible to neuroscientists who are not expert programmers, lowering the barrier to entry for state-of-the-art analysis.

K. Automated Analysis of Cellular Signals from Large-Scale Calcium Imaging Data (2009)

This is one of the first paper to analyse the basis of calcium imaging, which pioneered an automated framework for extracting cellular signals, from large-scale calcium imaging movies. The research points out to the limitations of manual region-of-interest (ROI) methods that are labour-intensive and inaccurate due to human error. The proposed method was a pipeline which integrates Principal Component Analysis (PCA) for dimensionality reduction followed by Independent Component Analysis (ICA) to separate statistically independent cellular sources based on spatial and temporal sparseness. The study demonstrates the ability to distinguish and even extract activity from the hundreds of densely neurons and glia, which are stable across the behavioural states and revealing fine –scale functional microzones . This could be considered as the basis of calcium imaging as it works on scalable analysis of complex calcium imaging datasets, catalysing future algorithmic developments in neural signal demixing. The dataset used in this study was utilized synthetic calcium imaging data generated via simulations and in vivo data from anesthetized rats.

The performance metrics applied here were True Positive Rate (TPR), False Positive Rate (FPR) and Computational efficiency.

L. Long-term, high-resolution in vivo calcium imaging in pigeons. (2024)

This study informs about the groundbreaking method for performing in vivo two photon calcium imaging in awake pigeons, achieving long –term, high –resolution visualizations. The encoded calcium indicator **GCaMP6s**, was deployed to label the neurons. The, those labelled neurons were imaged over extended periods at both single-cell and single-dendrite resolution.

The research was considered an advancement in calcium imaging. It demonstrated longitudinal, subcellular imaging in pigeons, which further could have mapped and observed memory formation in birds, but due to limitations such as cell-type specificity, it guaranteed future extended research in this area. The dataset used in this study was in vivo two-photon calcium imaging of GCaMP6s-expressing neurons in the visual Wulst and hippocampus of awake pigeons.

Here the performance metrics were imaging resolution, signal stability, and neuronal activity detection.

M. Multi-region calcium imaging in freely behaving mice with ultra-compact head-mounted fluorescence microscopes (2023)

The peers in this study, used a tightly Integrated Neuronal Imaging microscope (TINI), which is ultra –compact, and weighs around 0.43 g. This lightweight machine was used in enabling simultaneous calcium imaging from up to four distinct brain regions in freely behaving mice, including subcortical sites. Their findings revealed the dynamic activity patterns distributed across these interconnected areas. It was a step towards, multimodal circuit level investigations in freely moving animals.

The authors contributed to multi-region neural imaging, overcoming limitations related to weight, size, and head-space constraints for calcium imaging. The dataset was extracted using the TINI microscope to record calcium signals from up to four brain regions in freely behaving mice. Here the performance metrics were neuron detection, Signal-to-noise Ratio (SNR), and spatial resolution.

N. Spatial Redundancy Transformer for Self-Supervised Fluorescence Image Denoising (2023)

The paper introduces SRDTrans , which is a self-supervised denoising method. Traditionally, a reference image is taken which is the ground truth, which is therefore worked on with. SRDTrans employes spatial redundancy strategy, it splits a single image with noise into multiple overlapping images, to create training pairs for the network.

The authors have used a lightweight transformer network that can be used to capture long – range patterns, thus even the fine details in the images, which maintained low computational costs. This made it a convenient technique for 3D fluorescence imaging data. Experiments show that SRDTrans **improves image quality** significantly by increasing the signal-to-noise ratio and restoring delicate details. It performs better at preserving the structures and features in the images. This method was phenomenally successful in denoising high–resolution fluorescence images. For the dataset, authors simulated noisy fluorescence images with varying levels of SNR to train and evaluate the model.

Here the performance metrics are Peak Signal-to-Noise ratio (PSNR), structural similarity index and again computational efficiency.

O. ICoRD: Iterative Correlation-Based ROI Detection Method for the Extraction of Neural Signals in Calcium Imaging (2022)

Now this paper proposes a different approach, it discusses about ICoRD , a method designed to extract neural signals from calcium imaging, especially when signal-to-noise ratio (SNR) is low. ROI detection in calcium imaging is a crucial process, which often cause struggle in noisy conditions, leading to poor signal extraction.

So, to tackle this challenge, ICoRD , used an iterative process, which works on refining the ROIs. It selects an initial reference signal from neighbouring pixels, then updates the reference to better match the true neural signal. The method was tested on simulated calcium imaging data with varying SNR levels and compared with existing techniques. Results showed that ICoRD consistently provided more accurate signal extraction, particularly in low SNR scenarios. The dataset employed both simulated calcium imaging data, with the conditions of a controlled noise environment.

Here the performance metrics were Signal Extraction Accuracy, True Positive Rate, and False Positive rate.

III. RESEARCH GAPS IDENTIFIED

A. Limited Long-term, high-resolution imaging

Long-term calcium imaging has been successful in mice and pigeons as discussed previously, but due to Montion artifacts, and imaging depth i.e., the clarity and no of sub-images used to construct a calcium imaging movie restricts these techniques in freely behaving animals

B. Insufficient Noise Methods

Traditional filtering and supervised learning, often require clean reference datasets, which are hard to obtain in vivo. Self-supervised approaches like SRDTrans are still in experimental stages, therefore, yet to be validated across diverse neural circuits and imaging conditions.

C. ROI Detection Challenges

In low SNR environments, methods like ICoRD could be employed but, due to its iterative and correlation-based approach, it becomes computationally intensive, which is not scalable for large volumetric dataset.

IV. Conclusion

Spatio-temporal denoising of calcium imaging data is critical for mapping of neural activity. This paper even algins with addressing the goals of AI4Life Calcium Imaging Denoising Challenge 2025. Advances in self-supervised denoising methods, iterative ROI detection, and lightweight imaging hardware have significantly improved signal fidelity and spatial resolution. Despite recent advancements , we face challenges in achieving robust and long – term imaging across multiple brain regions, under in vivo conditions.

We would propose an integration between existing techniques with deep learning methods, which would not only reduce the computational cost, but even deal with large datasets, learning to leverage noise characteristics and enhance signal quality without clean reference images.

This will become a novel idea proposed, and further would facilitate comprehensive mapping of neural circuits, the study of neural dynamics, and even enhance reliability of calcium imaging as a tool.

V. Future Scopes

In the future, calcium imaging denoising can be improved by studying multiple brain regions at once. This research might help in developing real-time deep learning denoising will help track neural activity in freely moving animals. Applying these methods to different species and behaviours, along with automated ROI detection, more accurate analysis

References

[A] Robbins, M., Christensen, C. N., Kaminski, C. F., & Zlatic, M. (2021). Calcium imaging analysis - how far have we come? In *F1000Research* (Vol. 10, p. 258). NLM (Medline). https://doi.org/10.12688/f1000research.51755.1

[B] Pnevmatikakis, E. A. (2019). Analysis pipelines for calcium imaging data. In *Current Opinion in Neurobiology* (Vol. 55, pp. 15–21). Elsevier Ltd. https://doi.org/10.1016/j.conb.2018.11.004

[C] Stringer, C., & Pachitariu, M. (2019). Computational processing of neural recordings from calcium imaging data. In *Current Opinion in Neurobiology* (Vol. 55, pp. 22–31). Elsevier Ltd. https://doi.org/10.1016/j.conb.2018.11.005

[D] Pnevmatikakis, E. A., Soudry, D., Gao, Y., Machado, T. A., Merel, J., Pfau, D., Reardon, T., Mu, Y., Lacefield, C., Yang, W., Ahrens, M., Bruno, R., Jessell, T. M., Peterka, D. S., Yuste, R., & Paninski, L. (2016). Simultaneous Denoising, Deconvolution, and Demixing of Calcium Imaging Data. *Neuron*, *89*(2), 285. https://doi.org/10.1016/j.neuron.2015.11.037

[E] Malik, W. Q., Schummers, J., Sur, M., & Brown, E. N. (2011). Denoising two-photon calcium imaging data. *PLoS ONE*, *6*(6). https://doi.org/10.1371/journal.pone.0020490

[F] Chaudhary, S., Moon, S., & Lu, H. (2022). Fast, efficient, and accurate neuro-imaging denoising via supervised deep learning. *Nature Communications*, *13*(1). https://doi.org/10.1038/s41467-022-32886-w

[G] Li, X., Zhang, G., Wu, J., Zhang, Y., Zhao, Z., Lin, X., Qiao, H., Xie, H., Wang, H., Fang, L., & Dai, Q. (2020). *Reinforcing neuron extraction and spike inference in calcium imaging using deep self-supervised learning*. https://doi.org/10.1101/2020.11.16.383984

[H] Kamran, S. A., Moghnieh, H., Hossain, K. F., Bartlett, A., Tavakkoli, A., Drumm, B. T., Sanders, K. M., & Baker, S. A. (2024). Automated denoising software for calcium imaging signals using deep learning. *Heliyon*, *10*(21). https://doi.org/10.1016/j.heliyon.2024.e39574

[I] Ribeiro, A. H., Ribeiro, M. H., Paixão, G. M. M., Oliveira, D. M., Gomes, P. R., Canazart, J. A., Ferreira, M. P. S., Andersson, C. R., Macfarlane, P. W., Wagner, M., Schön, T. B., & Ribeiro, A. L. P. (2020). Automatic diagnosis of the 12-lead ECG using a deep neural network. *Nature Communications*, *11*(1). https://doi.org/10.1038/s41467-020-15432-4

[J] Cantu, D. A., Wang, B., Gongwer, M. W., He, C. X., Goel, A., Suresh, A., Kourdougli, N., Arroyo, E. D., Zeiger, W., & Portera-Cailliau, C. (2020). EZcalcium: Open-Source Toolbox for Analysis of Calcium Imaging Data. *Frontiers in Neural Circuits*, *14*. https://doi.org/10.3389/fncir.2020.00025

[K] Mukamel, E.A., Nimmerjahn, A. and Schnitzer, M.J., 2009. Automated analysis of cellular signals from large-scale calcium imaging data. Neuron, 63(6), pp.747-760.

[L] Nimpf, S., Kaplan, H.S., Nordmann, G.C., Cushion, T. and Keays, D.A., 2024. Long-term, high-resolution in vivo calcium imaging in pigeons. Cell reports methods, 4(2).

[M] Xue, F., Li, F., Zhang, K.M., Ding, L., Wang, Y., Zhao, X., Xu, F., Zhang, D., Sun, M., Lau, P.M. and Zhu, Q., 2024. Multi-region calcium imaging in freely behaving mice with ultra-compact head-mounted fluorescence microscopes. National Science Review, 11(1), p.nwad294.

[N] Li, X., Hu, X., Chen, X., Fan, J., Zhao, Z., Wu, J., Wang, H. and Dai, Q., 2023. Spatial redundancy transformer for self-supervised fluorescence image denoising. Nature Computational Science, 3(12), pp.1067-1080.

[O] Kang, S., Park, J., Kim, K., Lim, S.H., Kim, S., Choi, J.H., Rah, J.C. and Choi, J.W., 2022. ICoRD: iterative correlation-based ROI detection method for the extraction of neural signals in calcium imaging. Journal of Neural Engineering, 19(4), p.046035.